

CHE 565 -- Homework 5

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Introduction

This homework was completed using Python with the Control Systems Library. The report uses the same block-diagram logic as the Simulink assignment, but the systems are constructed programmatically using transfer functions, state-space conversions, and `ct.interconnect`.

The cascade-control problems compare the response with and without the inner feedback loop. The first controller is treated as an ideal PI controller, while the inner controller is P-only. For the no-cascade case, the inner feedback is disconnected and the inner controller gain is set to 1.

$$G_{c,1}(s) = K_{c,1} \left(1 + \frac{1}{\tau_{I,1}s} \right)$$

$$G_{c,2}(s) = K_{c,2}$$

$$IAE = \int_0^T |r(t) - y(t)| dt$$

```
1 # P-only, but represented as a transfer function so ct.ss() works.
2 return ct.tf([Kc], [1])
3
4 Gc2 = p_only_controller(Kc2)
5 Gc2_blk = ct.ss(Gc2, name='Gc2', inputs='E2', outputs='Yc2')%
```

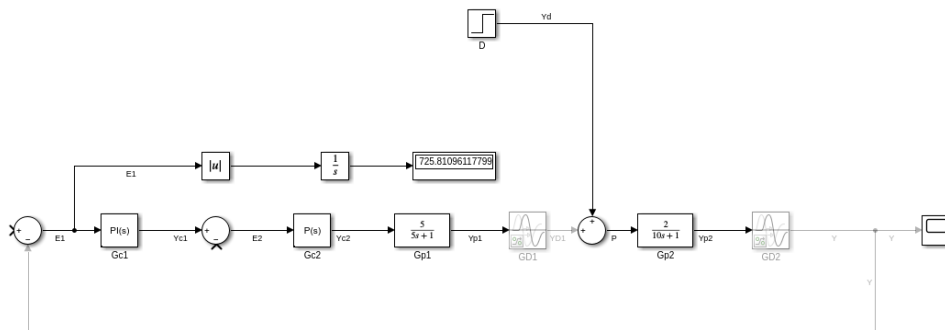


Figure 1: Simulink block diagram for the cascade-control system.

Problems 1--3: Cascade Control

Problem 1: System Without Cascade Control

For the no-cascade case, the inner feedback path was disconnected and the P-only inner controller gain was set to $K_{c,2} = 1$. The outer PI controller was then tuned using the autotuning from simulink and lambda tuning method and minimizing the IAE

$$K_{c,1} = 1.0000, \quad \tau_{I,1} = 1.0000, \quad I_1 = 1.0000$$

Case	$K_{c,1}$	$\tau_{I,1}$	IAE
Disturbance	1	1	450862.8604
Autotune	0.1837	12.2474	13.3271
Lambda	0.6094	4.3333	9.0895
IAE optimized	243.3753	32.1514	0.1404

Table 1: Problem 1 results without cascade control.

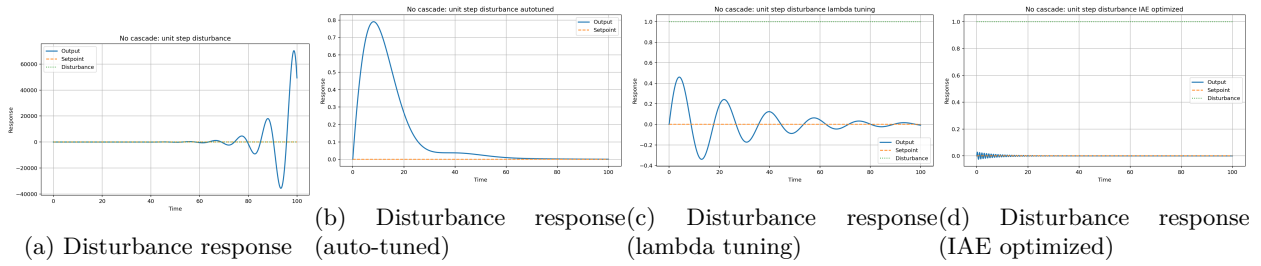


Figure 2: Problem 1 responses without cascade control.

Case	$K_{c,1}$	$\tau_{I,1}$	IAE
Setpoint	1	1	1233339.1635
Autotune	0.1837	12.2474	8.2136
Lambda	0.6094	4.3333	16.5618
IAE optimized	243.3753	32.1514	4.7511

Table 2: Problem 1 setpoint results without cascade control.

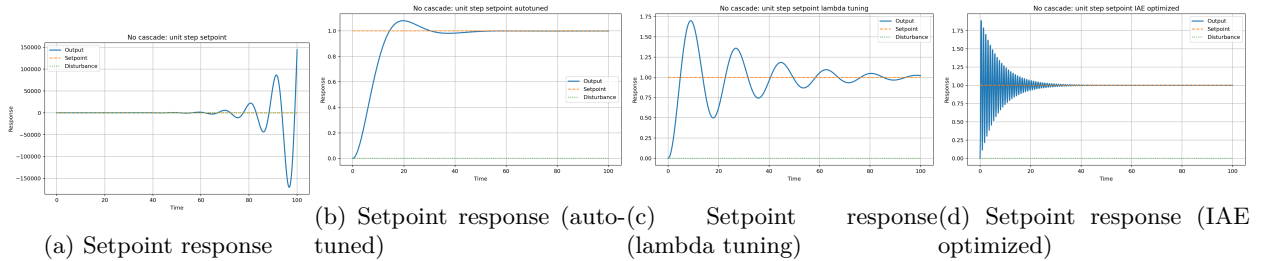


Figure 3: Problem 1 responses without cascade control.

Problem 2: System With Cascade Control

Problem 2: System With Cascade Control

For the cascade case, the inner feedback path was connected and the inner P-only controller gain was set to $K_{c,2} = 0.4$. The outer controller was tuned again because the outer loop sees a different equivalent process.

$$K_{c,1} = 1.0000, \quad \tau_{I,1} = 1.0000, \quad I_1 = 1.0000$$

Case	$K_{c,2}$	$K_{c,1}$	$\tau_{I,1}$	IAE
Disturbance	0.4000	1	1.0000	9.8851
Autotune	0.4000	0.7508	4.5461	4.1578
Lambda	0.4000	0.6094	4.3333	4.8484
IAE optimized	0.4000	2337.2940	953.8600	0.0441

Table 3: Problem 2 results with cascade control and unit step disturbance.

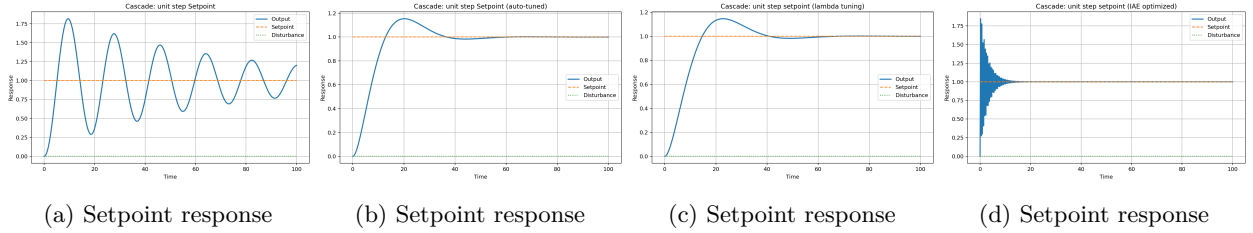


Figure 4: Problem 2 responses with cascade control and unit step setpoint changes.

Case	$K_{c,2}$	$K_{c,1}$	$\tau_{I,1}$	IAE
Setpoint	0.4000	1	1.0000	31.0631
Autotune	0.4000	0.7508	4.5461	8.7692
Lambda	0.4000	0.6094	4.3333	9.7764
IAE optimized	0.4000	2337.2940	953.8600	1.8478

Table 4: Problem 2 results with cascade control and unit step setpoint changes.

Problem 3: Comparison of Cascade and No-Cascade Results

Test	No Cascade IAE	Cascade IAE	Change
Disturbance	450862.8604	9.8851	-450852.9753
Setpoint	13.3271	4.1578	-9.1692

Table 5: Comparison of no-cascade and cascade-control performance.

Cascade control is expected to make the largest difference for disturbance rejection. The disturbance enters at the intermediate process variable, so the inner loop can react before the effect fully propagates to the final output. For setpoint tracking, cascade control may still change the response, but the benefit is usually less direct because the setpoint must pass through both the outer and inner loops.

Problem 4: Relative Gain Array for the 4 by 4 Process

$$\Lambda = K \circ (K^{-1})^T$$

$$K = \begin{bmatrix} 0.4300 & 0.4300 & 0.2300 & 0.2200 \\ -0.3300 & 0.3200 & -0.2000 & 0.2000 \\ 0.2200 & 0.2300 & 0.4200 & 0.4100 \\ -0.2200 & 0.2200 & -0.3200 & 0.3200 \end{bmatrix}$$

$$\Lambda = \begin{bmatrix} 0.6795 & 0.7167 & -0.1958 & -0.2004 \\ 0.8768 & 0.8566 & -0.3548 & -0.3786 \\ -0.1885 & -0.2078 & 0.6999 & 0.6964 \\ -0.3678 & -0.3654 & 0.8507 & 0.8826 \end{bmatrix}$$

y_i	u_1	u_2	u_3	u_4
y1	0.6795	0.7167	-0.1958	-0.2004
y2	0.8768	0.8566	-0.3548	-0.3786
y3	-0.1885	-0.2078	0.6999	0.6964
y4	-0.3678	-0.3654	0.8507	0.8826

Table 6: Relative gain array for Problem 4.

Output	Manipulated Variable	Λ_{ij}	$ \Lambda_{ij} - 1 $
y1	u2	0.7167	0.2833
y2	u1	0.8768	0.1232
y3	u3	0.6999	0.3001
y4	u4	0.8826	0.1174

Table 7: Suggested pairings for Problem 4.

The suggested pairings are chosen from positive RGA values that are relatively close to one while avoiding negative pairings. Negative relative gains are avoided because they indicate that closing other loops can reverse the apparent gain direction.

Problem 5: Relative Gain Array for the 2 by 2 Process

$$K = \begin{bmatrix} 5 & 2 \\ 3 & 6 \end{bmatrix}$$

$$\Lambda = \begin{bmatrix} 1.2500 & -0.2500 \\ -0.2500 & 1.2500 \end{bmatrix}$$

y_i	u_j	Λ_{ij}	$ \Lambda_{ij} - 1 $
y1	u1	1.2500	0.2500
y2	u2	1.2500	0.2500

Table 8: Relative gain array for Problem 5.

The diagonal pairings are preferred because the diagonal RGA values are positive and greater than one, while the off-diagonal values are negative. Therefore, the best pairing is y1 with u1 and y2 with u2.

Problems 6--10: MIMO Control, Cross Terms, and Decoupling

Problem 6: Two Single-Loop Controllers with Cross Terms Neglected

$$G_{11}(s) = \frac{5e^{-5s}}{4s+1} \quad G_{12}(s) = \frac{2e^{-4s}}{8s+1} \quad G_{21}(s) = \frac{3e^{-3s}}{12s+1} \quad G_{22}(s) = \frac{6e^{-3s}}{10s+1}$$

$$D_{12}(s) = \frac{2(4s+1)e^s}{5(8s+1)} \quad D_{21}(s) = \frac{10s+1}{2(12s+1)}$$

Loop	K_p	τ_p	θ	λ	K_c	τ_I
G11	5.0000	4.0000	5.0000	5.0000	0.0800	4.0000
G12	2.0000	8.0000	4.0000	4.0000	0.5000	8.0000
G21	3.0000	12.0000	3.0000	4.0000	0.9796	12.0000
G22	6.0000	10.0000	3.0000	3.0000	0.2778	10.0000

Table 9: PI tuning values for the two diagonal loops.

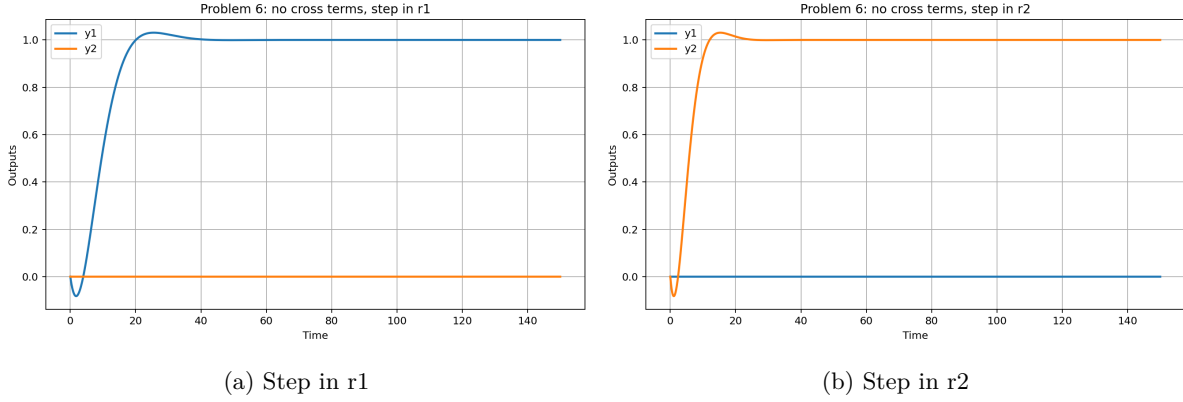


Figure 5: Problem 6 responses with cross terms neglected.

Problem 7: Cross Terms Included

Test	Controlled-output IAE	Interaction area
Step in r1	12.4992	4.5391
Step in r2	7.4995	4.3600

Table 10: Problem 7 performance with cross terms included.

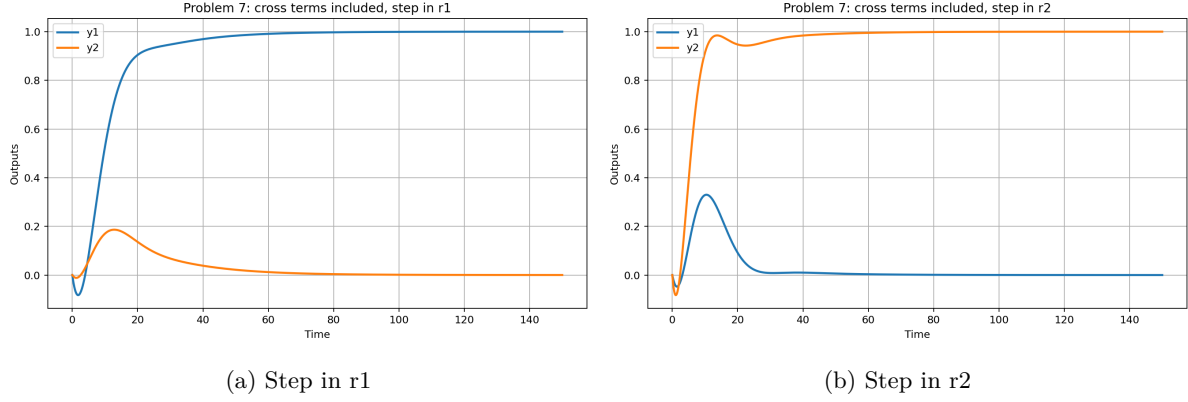


Figure 6: Problem 7 responses with cross terms included.

Problem 8: Static Decouplers

$$D_{12} = -\frac{K_{12}}{K_{11}} = -\frac{2}{5}, \quad D_{21} = -\frac{K_{21}}{K_{22}} = -\frac{3}{6}$$

Test	Controlled-output IAE	Interaction area
Step in r1	12.4998	0.6320
Step in r2	7.4998	2.1539

Table 11: Problem 8 performance with static decouplers.

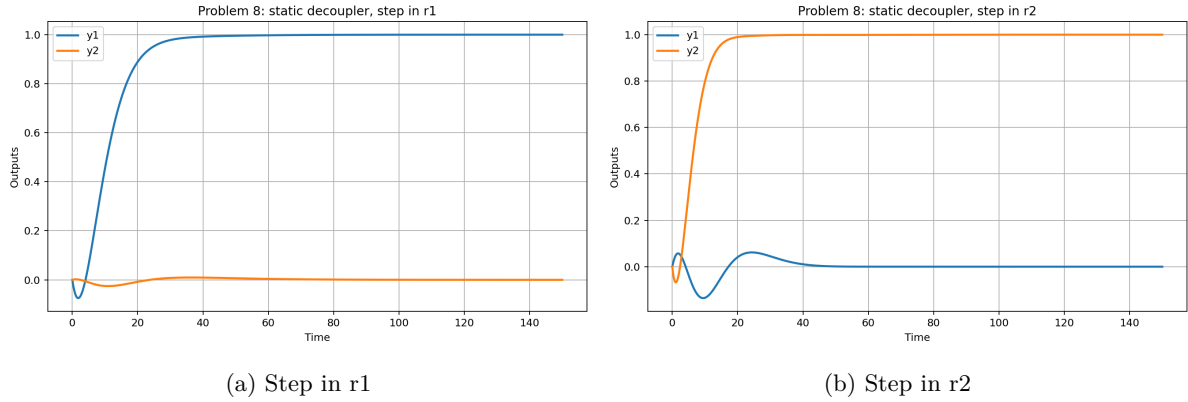


Figure 7: Problem 8 responses with static decouplers.

Problem 9: Dynamic Decouplers

$$D_{12}(s) = -\frac{G_{12}}{G_{11}} = -\frac{2}{5} \frac{4s+1}{8s+1} e^s$$

$$D_{21}(s) = -\frac{G_{21}}{G_{22}} = -\frac{3}{6} \frac{10s+1}{12s+1}$$

The term e^s in D_{12} is noncausal because it is equivalent to a negative delay. Therefore, the realizable dynamic decoupler uses only the rational part of D_{12} .

$$D_{12,realizable}(s) = -\frac{2}{5} \frac{4s + 1}{8s + 1}$$

Test	Controlled-output IAE	Interaction area
Step in r1	12.4998	0.0000
Step in r2	7.4998	0.6073

Table 12: Problem 9 performance with dynamic decouplers.

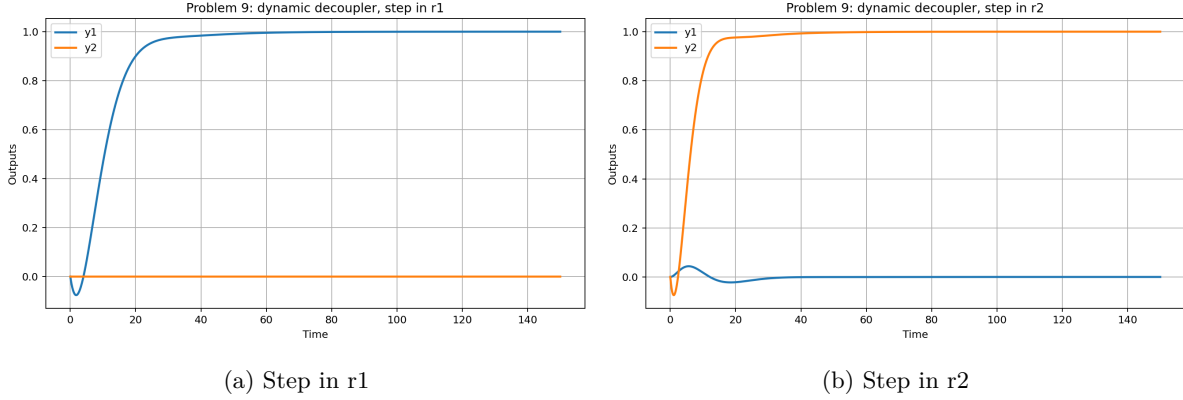


Figure 8: Problem 9 responses with dynamic decouplers.

Problem 10: Comments on MIMO Results

Case	Step in r1: IAE y1	Step in r1: y2 area	Step in r2: y1 area	Step in r2: IAE y2
No cross terms	10.7171	0.0000	0.0000	6.4302
Cross terms	12.4992	4.5391	4.3600	7.4995
Static decoupler	12.4998	0.6320	2.1539	7.4998
Dynamic decoupler	12.4998	0.0000	0.6073	7.4998

Table 13: Summary of MIMO response metrics for Problems 6--10.

When the cross terms are added, a change in one loop causes movement in the other output, showing loop interaction. Static decouplers reduce the steady-state interaction because they cancel the cross-gains at zero frequency. However, static decoupling cannot remove all transient interaction because the cross terms have different delays and time constants. Dynamic decouplers account for those dynamics and should reduce interaction further, but any noncausal negative-delay terms must be removed or approximated with a realizable transfer function.